JASHWANTH S

Data Scientist | AI Engineer | Data Engineer

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Professional Summary

Data Scientist and AI Engineer with hands-on experience building time-series forecasting, classification, and LLM-powered systems on top of scalable data pipelines. Proficient in Python, SQL, Pandas, NumPy, Scikit-learn, and TensorFlow, with expertise in predictive modeling (DeepAR), data validation and quality checks, and workflow orchestration (Airflow, Jenkins). Proven track record automating end-to-end analytics processes in enterprise environments — including cutting a monthly platform refresh across 200+ dashboards from 10 days to 2 — and delivering reliable results to business teams.

Technical Skills

Languages: Python, SQL, R, JavaScript

Data Science & Forecasting: Time-Series Forecasting (DeepAR), Classification, Predictive Modeling, Feature Engineering, Model Evaluation, Data Validation & QA

ML / DL Frameworks: TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras

LLMs & Agents: LangChain, HuggingFace Transformers, LLM Fine-tuning, RAG, AI Agents, Ollama, SentencePiece, FAISS, Prompt Engineering

Computer Vision & NLP: OpenCV, CNN architectures, YOLO (v8/v10), Transfer Learning, Multi-model Fusion, NLTK, BLEU/ROUGE evaluation

MLOps & Cloud: Docker, AWS, GCP, SageMaker, FastAPI, Streamlit, Apache Airflow, Apache Spark, Jenkins

Data & Databases: Pandas, NumPy, MySQL, PostgreSQL, MongoDB, Elasticsearch

Specializations: Generative AI, Agentic Workflows, Workflow Orchestration, Embedding Techniques

Work Experience

AI Engineer | Global Data — Aug 2025 – Present

- Built an end-to-end orchestration platform for the **Monthly Refresh (MR)** that keeps 200+ country-category dashboards current, covering vendor data ingestion, enrichment, Elasticsearch indexing, trend metrics (CAGR, engagement score), **DeepAR time-series forecasting** on social and video data, and classification; **cut the monthly cycle from 10 days to 2 days**.
- Replaced manual VM- and command-driven runs across multiple repos with Jenkins jobs and an Airflow DAG that uses a shared database as job state, checking it every 10 seconds to trigger dependent jobs, with built-in validation and QA at every stage.
- Added a natural-language control layer integrated with Slack, letting teams trigger runs, track status, receive error alerts, and resolve issues directly in chat; backed by a custom orchestrator UI and FastAPI backend.
- Built LLM-powered automated tagging workflows integrated with Spark, combining NER, TF-IDF/KeyBERT keyword extraction, and embedding-based semantic search to tag content at scale; used zero/few-shot LLM prompting via API for context-aware classification, reducing manual tagging effort by 70%.
- Automated data pipelines with Apache Airflow, orchestrating multi-step DAGs for ingestion, transformation, and validation across 1M+ rows weekly; incorporated retry logic, alerting, and dependency management to ensure fault-tolerant execution, cutting manual effort by 60%.
- Developed Spark-based enrichment pipelines handling 100GB+ datasets, applying distributed joins and custom UDFs to enrich records against internally validated mapping files and LLM-generated annotations, ensuring consistent, high-quality feature enrichment at scale.
- Built a data collection pipeline harvesting Google and YouTube search data using Playwright for browser automation with header caching and intelligent request throttling to avoid API rate limits; integrated LLM-based content classification and relevance scoring, with data stored in S3 and PostgreSQL, improving ingestion speed by 40%.

Data Engineering Intern | Ai Palette — Aug 2024 – Apr 2025

- Managed 10M+ records across production datasets ensuring high availability; reduced data retrieval errors by 25%.
- Engineered optimized real-time processing workflows increasing system throughput by 35% and improving response times by 20%.

- Built scalable pipelines automating data movement to production, cutting manual processing time by 40% and enabling near real-time analytics.

Key Projects

Predictive Maintenance for IoT Devices Jan 2025 – Mar 2025

- Engineered an ML system using Python, Scikit-learn, and TensorFlow to forecast IoT device failures with 92% accuracy and 87% recall across 500 simulated devices.
- Applied advanced feature extraction (rolling averages, FFT) and ensemble models (Random Forest, XGBoost, LSTM), achieving 15% improvement in recall over baseline.
- Deployed models via Flask on AWS with 1.2s average response time; built Streamlit dashboards processing 10,000+ data points per device.

Agentic RAG System Oct 2023 – Dec 2023

- Designed and built a Retrieval-Augmented Generation (RAG) agent using Python, SentencePiece, and FAISS to process 5,000+ documents with 2.3s average retrieval time and 85% relevance accuracy.
- Integrated BLEU, ROUGE, and BLEURT evaluation metrics achieving BLEU 0.34 and ROUGE-L 0.42 across 500 test QA pairs — establishing a rigorous LLM output evaluation pipeline.
- Deployed end-to-end on Streamlit + HuggingFace Transformers supporting 10+ concurrent users with 95% uptime.

SpeakEasy: Real-Time Indian Language Translation Jul 2024 – Dec 2024

- Fine-tuned transformer models on 100,000 parallel sentence pairs across 6 Indian languages, achieving 23.4 BLEU (English-Hindi) and 21.8 BLEU (English-Tamil).
- Delivered real-time translations at 1.8s latency per sentence with batch processing support via HuggingFace Transformers CLI and web interface.

Waste Management System – CNN Object Detection Oct 2023 – Dec 2023

- Trained a CNN to classify 10 waste material categories with 92% accuracy on 15,000 images; augmented dataset 3× to improve cross-validation accuracy from 87% to 92%.
- Integrated with OpenCV for real-time inference at 150ms per image with 90% precision. Patent filed: 202341082049 A.

Education

BE (Honors) — Artificial Intelligence & Machine Learning Aug 2025
Visvesvaraya Technological University

GPA: 9.53 CGPA | Ranked 10th in AIML Department | Award presented by Principal Secretary to the Government of Karnataka

GATE 2026 — DA Qualified Feb 2026
Score: 476 | Data Science and AI

Certifications

- Explore Generative AI with the Gemini API in Vertex AI — Google Cloud Jun 2025
- Inspect Rich Documents with Gemini Multimodality & Multimodal RAG — Google Cloud Jun 2025
- Develop GenAI Apps with Gemini and Streamlit — Google Cloud Jun 2025
- Data Analytics Job Simulation — Quantium Mar 2025
- Data Science Job Simulation — BCG Mar 2025
- Machine Learning with Python — IBM Oct 2023